

KUMUDH RANASINGHE

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Mount Lavinia, Colombo

PROFILE

I am an undergraduate with a strong passion for leveraging computer science to develop innovative, practical solutions. I am driven by the challenge of tackling complex problems and constantly seek smarter, more efficient approaches. With excellent self-management skills, I thrive under pressure, adapt to dynamic environments, and stay proactive in exploring emerging technologies to seamlessly integrate them into real-world applications.

EXPERIENCE

- Coachello** June 2025 - Present
Software Engineer (Freelance) Remote
 - Research and development efforts on the Coachello AI, contributing to the design and enhancement of AI features.
 - Full-stack development of the Coachello application to ensure a seamless user experience and robust performance.
 - Integrate and maintain third-party AI tools, ensuring compatibility with Coachello's infrastructure and optimizing for performance and scalability.
- Coachello** December 2024 - May 2025
Intern Software Engineer Remote
 - Full stack development and maintenance of the Coachello application, contributing to frontend (React, Gatsby, Tailwind), backend (FastAPI, MySQL) and AI (OpenAI, Anthropic, Gemini) development.
 - Optimize backend APIs for performance, reliability, and security, contributing to reduction in latency and downtime.
 - Assisted in UI/UX development, incorporating feedback from users to improve usability and user engagement.
 - Collaborated on the development and testing of the Coachello AI, helping to improve the user experience.

EDUCATION

- University of Moratuwa** 2022 - Present
BSc Eng Hons, Computer Science & Engineering CGPA - 3.7518
Specialization on Data Science & Engineering
- S. Thomas' College** 2018 - 2020
G.C.E. (A/L) Examination - 3A Passes (Physical Science Stream) Z-score - 2.3298
- Vidura College** 2007 - 2017
G.C.E. (O/L) Examination - 9A Passes

PROJECTS

- FYP - Vision Transformer based Pan-Arctic Feature Detection with Data Fusion** June 2025 - Present
Tools: Python, Detectron, QGIS, PyTorch, Rastario, OpenCV, Tensorboard
 - A unified ViT-CNN hybrid backbone for enhanced local-global feature extraction with context-aware patchification.
 - Leveraging Digital Elevation Models to improve feature detection accuracy by incorporating elevation data.
 - Enhancing the downstream segmentation heads by incorporating techniques such as deep-supervision and attention.
 - Deliverable of QGIS plugin, enabling real-world Arctic permafrost feature mapping for geospatial analysts.
- Quantum Trader - Algorithmic Trading Tool** November 2025 - Present
Tools: Python, PostgreSQL, Redis, PyTorch, StableBaselines3, MLFlow, RabbitMQ, Celery
 - A system combining Deep Learning, Reinforcement Learning, and XGBoost for automated signal generation.
 - Market regime detection and dynamic risk management, adjusting strategies based on volatility and model confidence.
 - Containerized with Docker, leveraging Celery for asynchronous task execution, RabbitMQ for message brokering, Redis for real-time caching, paired with MLFlow for experiment tracking and PostgreSQL for persistent signal storage.
- SatMAE-AdaFuse** August 2025 - October 2025
Tools: Python, Pytorch, Rastario, Timm, W&B
 - Enhanced satellite imagery pre-training with self-supervised learning for temporal and spectral fusion.
 - Introduced adaptive temporal sequencing and multi-head attention mechanisms for improved model performance.
 - Achieved significant accuracy improvements in land cover classification and remote sensing tasks.

• **ProjecX - Agentic Project Management Tool**

March 2025 - June 2025

Tools: FastAPI, NextJS, PostgreSQL, Azure, Vercel, OpenAI

- AI-powered project management tool to automate task creation, documentation, and resource allocation.
- Leveraged AI agents for task tracking and insights generation based on project progress and performance.
- Incorporate advanced decision support to enhance operational efficiency and minimize manual interventions.

• **OboTutor - AI Tutoring Chatbot**

June 2024 - November 2024

Tools: Django, React, MySQL, ChromaDB, Langchain, OpenAI, AWS

- AI-driven tutoring chatbot for personalized student education using LLMs, RAG, prompt engineering techniques.
- Curriculum-specific guidance with adaptive learning based on each student's progress and preferences.
- WhatsApp integration for easy student interaction, maintaining conversation history and resource recommendation.

• **E-commerce Microservices Polyglot**

July 2025 - August 2025

Tools: FastAPI, Golang, ExpressJS, PostgreSQL, Redis, Docker, Kubernetes

- E-commerce system with separate microservices for user authentication, product catalog, and order processing.
- Utilized polyglot programming with Python, Go, and Javascript for efficient service development.
- System containerized with Docker and orchestrated deployments using Kubernetes for scalability and reliability.

• **More Projects**

- AI Resume Builder - NextJS, Prisma, Clerk, TailwindCSS February 2025 - March 2025
- Mobile Weather Application - Flutter, OpenWeatherMap December 2024 - January 2025
- Stock Forecasting Tool - React, Flask, PyTorch, Scikit-Learn June 2024
- Resturent Application - React, ExpressJS, MongoDB, Stripe May 2024
- RPAL Interpreter - Python March 2024 - June 2024
- IOT wearable for Workplace Safety - Arduino, Node-Red, MQTT, NodeJS December 2023 - February 2024
- Airline Reservation System - React, ExpressJS, MySQL August 2023 - November 2023

SKILLS

- **Programming Languages:** Python, Java, C++, Javascript, Golang, SQL
- **Database Systems:** MySQL, PostgreSQL, MongoDB, Firestore, Redis, AstraDB
- **Tools & Services:** Azure, AWS, Vercel, Git, Jira, Docker, Kubernetes, MLFlow, RabbitMQ
- **Frameworks:** FastAPI, Flask, Django, Streamlit, NodeJS, ExpressJS, React, NextJS, Flutter
- **Machine Learning:** Pandas, Scikit-Learn, Pytorch, OpenCV, Tensorflow, Cuda, Detectron
- **Artificial Intelligence:** LLMs (OpenAI, Gemini, etc..), Frameworks (Langchain, LlamaIndex), RAG
- **Technical Fields:** ML/DL, DS, NLP, CV, GenAI, Full Stack Development

CERTIFICATIONS

- **AWS:** Machine Learning Foundations, Intro to Generative AI, Getting Started with Security, Intro to Cloud 101
- **Kaggle:** Intro to Machine Learning, Intermediate Machine Learning, Machine Learning Explainability, Intro to Deep Learning, Computer Vision, Advanced SQL
- **Nvidia:** Getting Started with Accelerated Computing in CUDA C/C++, Getting Started with Deep Learning, Fundamentals of Deep Learning
- **CIMA:** Certificate in Business Accounting (Cert BA)

REFERENCES

• **Dr. Thanuja D. Ambegoda** (Final Year Project Supervisor)

Department of Computer Science and Engineering
University of Moratuwa

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• **Dr. Amal S. Perera** (Final Year Project External Supervisor)

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• **Prof. Dulani Meedeniya** (Senior Lecturer)

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• **Mr. Dinal Kurukulasooriya** (Internship Supervisor)

CTO/Co-founder
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